Joyvishek Ghosh

Module Lead — AI Architect

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PROFILE SUMMARY

AI Architect and Module Lead with 8+ years of product engineering experience. Architects the enterprise AI layer at ARC Document Solutions: LLM tool-calling orchestration, GraphRAG over Neo4j and Qdrant, long-term user memory, a voice assistant endpoint and guarded, grounded answers over real facility data, while leading a 12–15 member engineering team. Focused on turning AI prototypes into secure, grounded, production-grade systems that work over real enterprise data.

TECHNICAL SKILLS

Generative AI	LLM Orchestration, Agentic AI, Tool Calling, RAG, GraphRAG, Prompt Engineering, AI Guardrails, Claude (Anthropic API, AWS Bedrock), Gemini, OpenAI, Ollama, mem0, Rasa
Vision / Doc AI	YOLOv8/v11, Object Detection, Instance Segmentation, VLMs, OCR (Surya, Tesseract), OpenCV
Data / Retrieval	Neo4j (Cypher), Qdrant, pgvector, PostgreSQL, MongoDB, Redis, ClickHouse, Elasticsearch, Solr
Backend	Python, FastAPI, Node.js, NestJS, Docker, Microservices, Event-driven Architecture, gRPC, Socket.IO
Cloud / APIs	AWS (Bedrock, S3, Secrets Manager), Google Maps Platform, Google Custom Search, YouTube Data API
Frontend	React.js, Next.js, Angular, TypeScript, Micro-frontends, PSPDFKit, PDF.js

WORK EXPERIENCE

Module Lead — AI Architect

Sep 2024 – Present

ARC Document Solutions, Kolkata

Enterprise AI Assistant & Orchestration

- Architected **ARC Facilities AI**, an enterprise assistant that answers natural-language questions over campus buildings, assets, pins and locations, with every answer backed by graph data, vector search, internal APIs or user memory.
- Built the Claude tool-calling loop that routes each request across graph Q&A, Google Maps location search, Gemini-grounded web search, Grainger industrial supplies, YouTube, uploaded-file reading and data export and email, with separate tool sets for single-campus and multi-campus sessions.
- Built the export and email tools to resolve “export this” against the last exportable answer in history, delivering files as S3 presigned links that expire and re-presign on click.
- Built the Redis-backed session layer for chat history, preferences and session state, with paged history and newline-delimited JSON streaming to clients.

GraphRAG, Memory & Voice

- Designed the **GraphRAG** pipeline: Qdrant supplies canonical entity, few-shot Cypher and past-failure hints, and two competing Cypher-generation chains (asset-first and pin-first) run in parallel against Neo4j, with every query campus-scoped.
- Added a bounded Cypher self-repair loop that feeds Neo4j errors back to the model, and rejects repaired queries that drop the campus filter or return hard-coded literals instead of data.
- Built post-answer grounding verification and anti-example recording, so silent graph-query failures are measured and reusable as negative examples.
- Added long-term user memory on mem0 with pgvector, with detect, confirm, override and delete flows for preferences, and a live-classifier benchmark covering English, Spanish and Tagalog.
- Built the voice assistant endpoint (*/campusmetadata*) on the same tool-calling loop and memory layer as text chat.
- Built fuzzy campus and building-name matching (RapidFuzz) so natural-language place names resolve to campus records.
- Built per-query token-usage tracking that normalizes response shapes across Anthropic, Bedrock and Converse, cutting cost on repeat questions through prompt caching and history trimming.

Data, Safety & Model Routing

- Operated the data layer across Redis (sessions, history, caches), Neo4j, Qdrant, pgvector, Elasticsearch (support knowledge base) and ClickHouse (analytics), with Claude served through AWS Bedrock or the first-party API behind a single kill switch.
- Built a three-layer content guard: prompt-injection filtering, an internals-extraction guard with an output sanitizer, and a multilingual hate and abuse filter that screens English, Spanish and Tagalog.

Skysite Intelligence: AI Document Intelligence

- Architected **Skysite Intelligence**, an AI document-intelligence layer for facility management that ingests engineering drawings and project documents and answers natural-language questions with citations back to the source sheet.
- Built a hexagonal, event-driven pipeline on Kafka: each stage (splitting, reading, knowledge graph, embeddings, retrieval) is an independent consumer, with domain logic kept separate from infrastructure adapters.
- Built drawing understanding that reads title blocks, keynotes, legends, schedules and egress elements, and measures room areas from the drawn walls, using PyMuPDF and OCR with a vision LLM.
- Built hybrid vector and graph retrieval with a reranker, and plans walking routes over the doorways in drawn walls.
- Kept inference fully self-hosted (vision LLM, embeddings and reranker on vLLM and TEI in production), with no calls to a third-party model vendor.

Computer Vision & Document AI

- Built OCR pipelines (Surya OCR, Tesseract, PDF text layers) for blueprints, technical manuals and facilities documents.
- Trained and deployed YOLO models for building annotation and equipment detection, and YOLOv11 instance segmentation for floor-area and structural-boundary isolation.
- Designed architectural-drawing intelligence that reads drawing sets (title blocks, legends, keynotes, schedules) using a VLM, knowledge graph and hybrid search, and cites the exact sheet in each answer.

Security & Production Readiness

- Led hardening before production rollout: closed authentication gaps, enforced user isolation, added prompt-injection guardrails, moved secrets from hard-coded values to AWS Secrets Manager with per-environment config, set input-size limits and per-user daily rate limits, and fixed dependency vulnerabilities.
- Added retries, fallbacks and structured logging so the assistant degrades gracefully when memory, analytics or external services are unavailable.

Product Engineering & Leadership

- Led a 12–15 member engineering team across requirements gathering, architecture, delivery and production support.
- Implemented role-based access control across Equipment, Compliance, Forms and Pin; led the Compliance 1.0 to 2.0 data migration and added dynamic surveys.
- Upgraded PSPDFKit in Forms (share links, report downloads, 1.0 to 2.0 conversion) and built a PDF viewer and signing tool with pdf-lib, PDF.js and Signature Pad.
- Built a Mutation Observer bridge for real-time React (host) and Angular (remote) micro-frontend communication; integrated Google Maps for campus location search.
- Built NestJS services (clean/onion architecture) with PostgreSQL and Redis: Equipment bulk upload, Forms activity tracking, recurring task scheduling and email notifications.

Full Stack Developer Jun 2023 – Sep 2024

Trigent Software (client: ARC Document Solutions)

- Built client and server applications with React, Next.js, Redux, NestJS, Node.js, PostgreSQL and MongoDB, including YOLOv8 computer-vision features.
- Shipped features that drove a **20% increase in user engagement**; resolved 15% more technical issues within deadlines and improved project completion time by **30%** through agile practices.
- Integrated Redis, Socket.IO, Solr and Elasticsearch; used Moleculer microservices for job/worker nodes and Turbo.js for micro-frontends.

Artificial Intelligence Consultant Feb 2021 – Mar 2023

Pemari Technology Pvt. Ltd.

- Built a Rasa-powered conversational AI chatbot for Broadcom's Clarity PPM, with a React front end and a Meteor/Node.js admin panel.
- Designed rich chatbot responses: tables, charts, images, video, guided flows, ratings, buttons and carousels.
- Connected the admin panel to Rasa through GraphQL/Apollo Server; added multitenancy for per-client data isolation and multiprocessing for performance.
- Set up automated Docker image builds into a private registry; led delivery on a 6-person team.

Full Stack Developer Mar 2019 – May 2019

Fleapo Corporation

- Built web applications including Realbooks and a T-shirt e-commerce site with Angular, Node.js and MongoDB.
- Worked directly with customers on progress and feedback; reviewed code, debugged issues and shipped fixes.

Full Stack Software Developer Mar 2017 – Mar 2019

TheoreX Edutech LLP

- Developed and maintained applications in AngularJS, Node.js, MongoDB, HTML5 and CSS3; reviewed code and debugged issues.

- Worked with clients across project stages and with engineers to improve software/hardware interfaces.

PERSONAL R&D PROJECTS

- **arch3D — Drawing to 3D/BIM Model.** Converts architectural drawings (PDF, image, DXF/DWG) into editable 3D/BIM models (GLB, IFC4) through a 12-stage pipeline: OCR, wall/door/window detection, scale resolution and room reconstruction, with confidence scores and human review. Geometry comes from measured data, never from the LLM. *Next.js*, *NestJS*, *FastAPI*, *OpenCV*, *Tesseract*, *IfcOpenShell*, *PostgreSQL*, *Redis*.
- **AutomationOS — AI Testing Platform.** QA automation platform that runs Excel-based test suites against an AI assistant, keeps multi-turn follow-up sessions, streams live results over SSE, and auto-validates responses with an LLM judge (Ollama) using configurable QA rules and similarity scores. Includes voice test automation via LiveKit and pass/fail and latency reports. *FastAPI*, *React*, *PostgreSQL*, *Redis*, *Celery*, *Docker*.
- **Cognitive Runtime (USCR).** Domain-agnostic runtime that turns plain-language questions into Neo4j, SQL or vector queries, driven by configurable ontology packs, with multi-tier memory and planning/reasoning agents. *FastAPI*, *Neo4j*, *Qdrant*, *Redis*.
- **AgentSync.** Protocol for AI agents to negotiate and complete real-world actions with secure sessions, policy checks, confirmation gates and audit logs. *FastAPI*, *React*, *Qdrant*, *Redis*, *Celery*.

EDUCATION

PG Diploma — International Institute of Information Technology, Bangalore 2023

B.Tech / B.E. — West Bengal University of Technology (WBUT) 2017

ISC (12th) / ICSE (10th) — CISCE 2013 / 2011

LANGUAGES

English, Hindi, Bengali